Rational Functions & Asymptotes

Vertical asymptotes and holes

1 Find the vertical asymptotes and holes of $f(x) = \dfrac{x^2 - 9}{x^2 - x - 12}$.

2 Find the y-coordinate of the hole in the function from the previous problem.

Horizontal and slant asymptotes

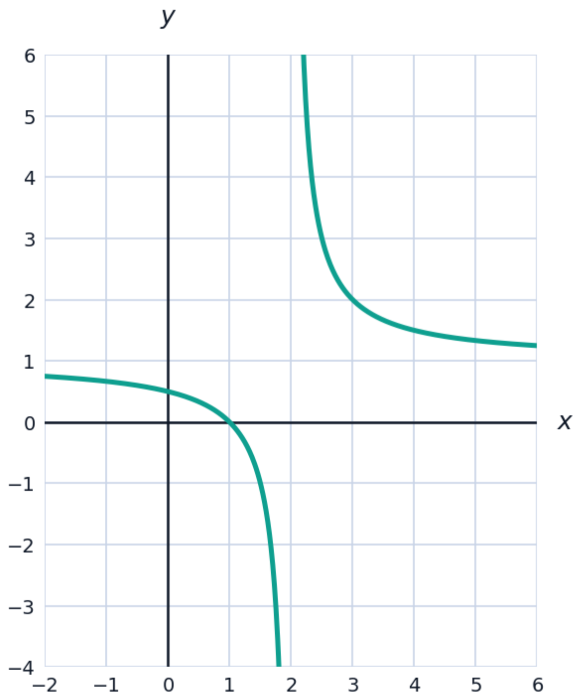
3 Find the horizontal asymptote of $f(x) = \dfrac{3x^2 - 2x + 1}{5x^2 + 4}$, using the degrees of numerator and denominator.

4 Find the horizontal asymptote of $g(x) = \dfrac{2x + 7}{x^2 - 1}$.

5 Explain why $h(x) = \dfrac{x^2 + 1}{x - 2}$ has a slant (oblique) asymptote instead of a horizontal one, and find it using polynomial division.

Visualizing a rational function's asymptotes

6 The graph shows $f(x) = \dfrac{1}{x - 2} + 1$, a transformed reciprocal function.



a Identify the vertical asymptote shown.

b Identify the horizontal asymptote shown.

End behavior of rational functions

7 Describe the end behavior of $f(x) = \dfrac{2x^3 - x}{x^2 + 1}$ as $x \to \pm\infty$ (note the numerator's degree exceeds the denominator's by 1, so the function behaves like its slant asymptote).

Domain and intercepts

8 Find the domain, x-intercept(s), and y-intercept of $f(x) = \dfrac{x - 5}{x^2 - 4}$.

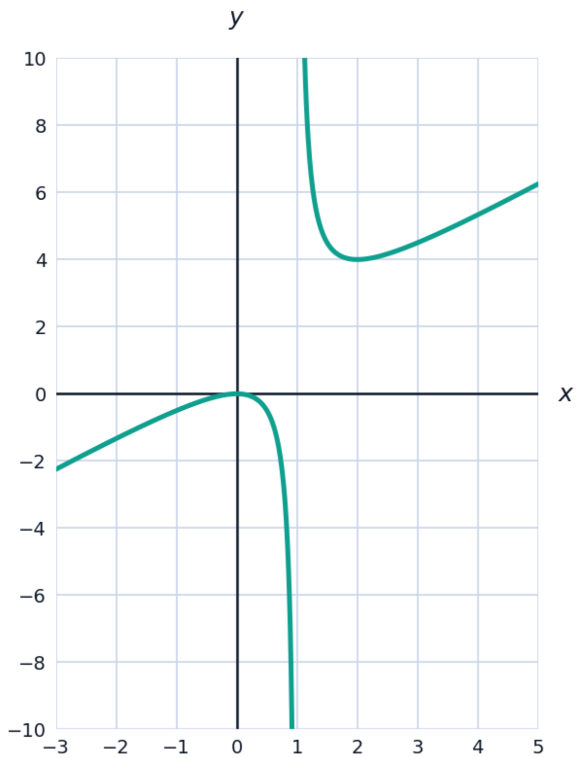
Solving rational equations and inequalities

9 Solve $\dfrac{x + 1}{x - 3} = 2$, checking that the solution doesn't create a zero denominator.

10 Solve the inequality $\dfrac{x - 1}{x + 2} > 0$ by determining the sign of the expression on each interval created by the zero and the excluded value.

Visualizing a function with a slant asymptote

11 The graph shows $f(x) = \dfrac{x^2}{x-1}$, which has a slant asymptote.



a Find the vertical asymptote and confirm it matches the graph.

b Use polynomial division to find the slant asymptote equation.

c Explain why this function has no horizontal asymptote.

Building a rational function from given features

12 Construct a rational function with a vertical asymptote at $x = 3$, a hole at $x = -1$, and a horizontal asymptote at $y = 2$.

13 Explain why a rational function cannot have both a horizontal asymptote and a slant asymptote at the same time.

Rational functions in context

14 The concentration of a drug in the bloodstream t hours after a dose is modeled by $C(t) = \dfrac{50t}{t^2+4}$ (mg/L). Find the horizontal asymptote and interpret it in context.